Bedfordview Primary School
TERM 4 Year End Test

Name: Memo	Grade 5 : A B C D
Subject: Natural Science and Technology	
Examiner: Mrs C. Nash	Date: November 2023
Moderator: Mr J. Nash	Time Allocation: 1hour 30min Total: 60
INSTRUCTIONS 1. Read all the instructions and questions carefully. 2. Write neatly and legibly. 3. Write in full sentences unless otherwise instructed. 4. Look carefully at the mark allocation when giving your answers. 5. Answer in the spaces provided. This paper consists of 6 pages.	Parent's signature & comments: _____ _____ _____

1	2	3	4	5	6	7
0 – 29%	30 – 39%	40 – 49%	50 – 59%	60 – 69%	70 – 79%	80 – 100%

	Section A: Term 3 Energy & Electricity and Energy & Movement							Section B: Term 4 Planet Earth and Surface of the Earth			
Question	1	2	3	4	5	6	7	1	2	**Total**	**%**
Allocated mark	**7**	**3**	**13**	**5**	**3**	**7**	**3**	**11**	**8**	**60**	
Learner's mark											
Moderated mark											

Content: **Energy and Electricity**: cells and batteries; mains electricity. **Energy and Movement**: wheel and axle; elastics and springs. **Planet Earth**: the earth moves and **Surface of the Earth**
Skills: Understanding and interpreting data on storing of energy, circuits, electricity and transferring of energy. Identifying objects that are elastic. Understanding stored energy and movement of energy

SECTION A – Energy and Electricity & Energy and Movement **/41**

Question 1: Definitions (Knowledge recall) **/7**

Match the words in column A with their meaning in column B.

Column A	**Column B**
A. Components	1. Things you do to stop something dangerous happening.
B. Rotation	2. The movement of an object in space around another object.
C. Precautions	3. Groups of related parts that work together as a whole for a particular purpose.
D. Systems	4. The parts that make up an electric circuit.
E. Revolution	5. Kinetic energy.
F. Compress	6. The movement of an object around itself.
G. Movement	7. To press or squash together.

A- 4	B- 6	C- 1	D- 3	E- 2	F- 7	G- 5

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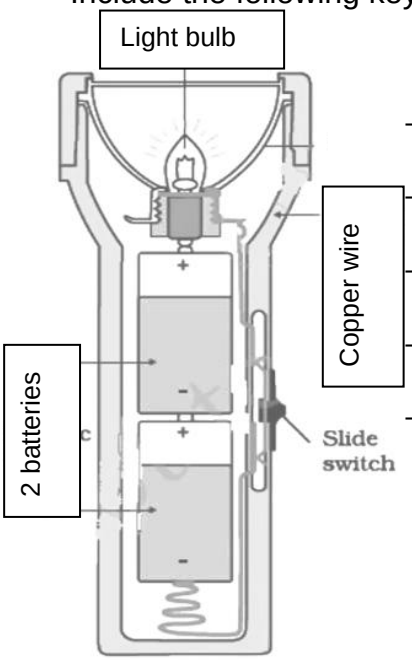
Questions 2: Cells and batteries (Data interpretation)

2.1 Describe how the electricity would flow through a torch.

Use the diagram with labels to guide you.

Include the following key words: **battery, copper wire, light bulb, switch** **/3**



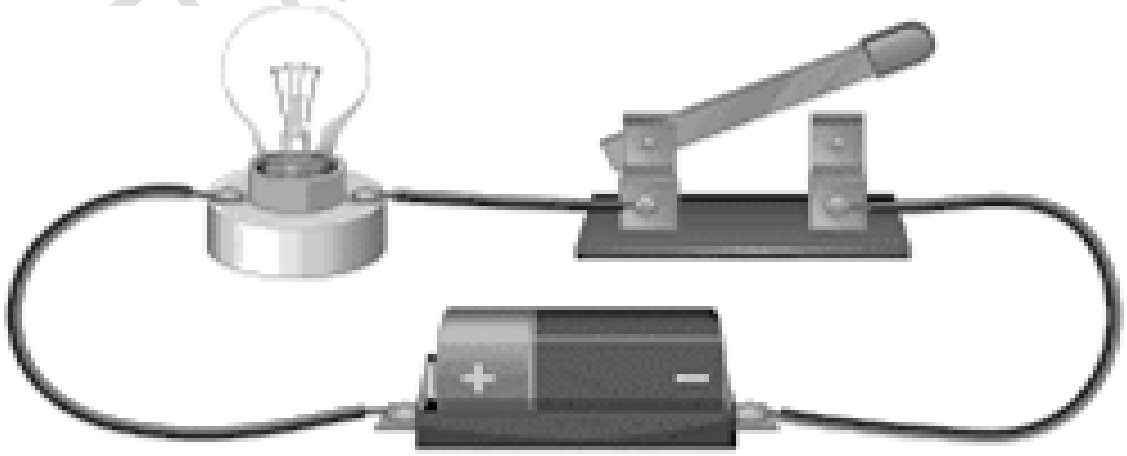
The battery provides the electrical energy to the circuit. This energy flows through the **copper wire**.

The **light bulb** will only shine when the **switch** is on and current is flowing

Questions 3: Electrical Circuit

Study the electrical circuit below and answer the questions that follow.

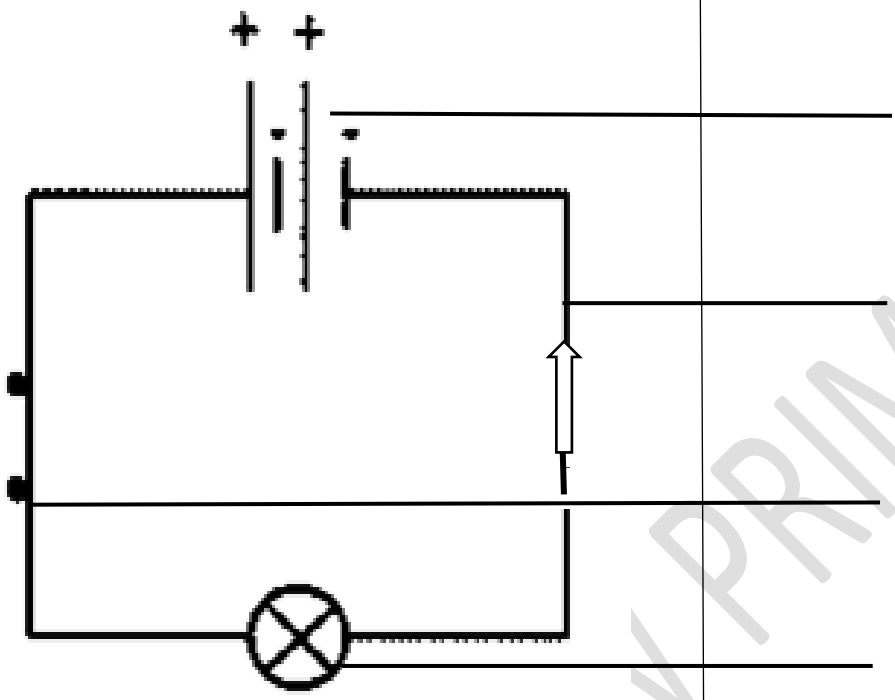
3.1 **Explain the function** of each component in the following electrical circuit: **/4**



Battery: Source of electrical energy
Switch: To connect and disconnect the power supply from battery to light bulb
Light bulb: To shine when the circuit is connected properly.

Copper wire: Transfer the flow of electrical energy to where it is needed.

3.2 **Draw and label** a closed **electrical circuit** with all the necessary **components** using the correct symbols in the space provided. **/8**

Diagram	**Labels**
	Cells - Battery Copper - wire Closed Switch Light Bulb

3.3 Predict what would happen if the copper wires are not properly connected to the electrical components? **/1**

Explain your reasoning.

The light bulb will not shine as there is no flow of electrical energy.

Question 4: **/5**

Draw a circle around the letter (A – D) to select the correct answer.

1.1 For a candle to keep on burning, the flame needs …

a) Oxygen
b) Carbon dioxide
c) Oxygen and Carbon dioxide
d) None of the above **/1**

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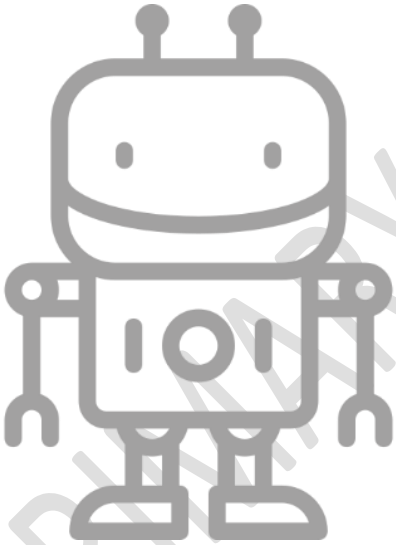
1.2 Stored energy can be changed into… energy in a toy car.

a) Light
b) Kinetic
c) Sound
d) Heat /1



1.3 An example of a useful output energy is…

a) Heat
b) Noise
c) Smoke
d) None of the above /1

1.4 To prevent a fire from spreading we should…

a) Always play with matches and lighters
b) Never leave a fire unattended at a camp site
c) Place wet clothes over a heater to dry
d) Stay indoors and make no attempt to get out /1

1.5 An important example of an everyday fuel is… (Two possible answers)

a) Petrol
b) Water
c) Oxygen
d) Food /1

Question 5: /3

Select the correct answer to make the following statements true.

1.1 A radio changes electrical energy into _____energy /1

a) Light
b) Heat
c) Movement
d) Sound

1.2 South Africa's main power station uses _____ /1

a) Water
b) Wind
c) Sun
d) Coal

1.3 The input energy for an electrical lightbulb is_____ /1

a) Light
b) Heat

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c) Electricity
d) Stored

Question 6: Mains electricity **/7**

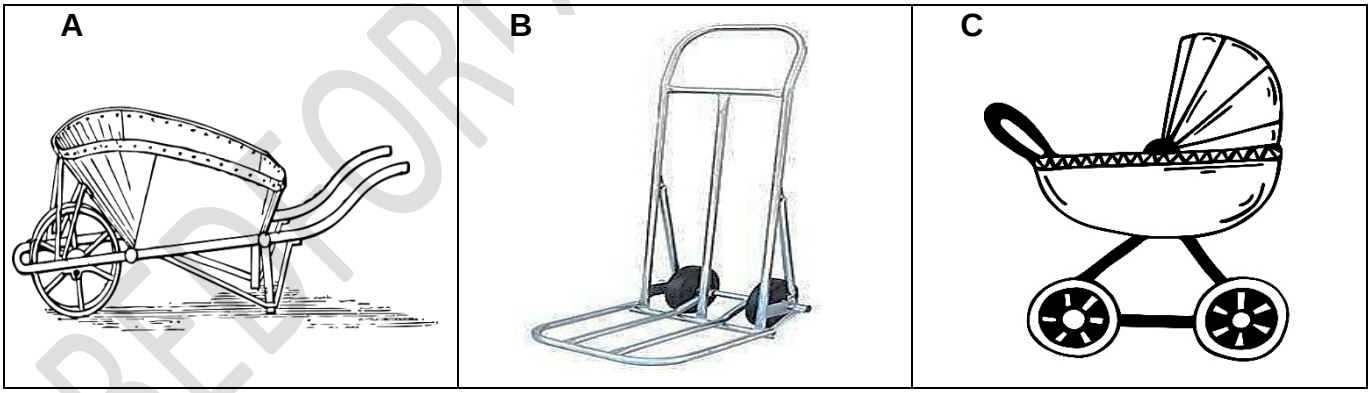
Place the following statements in their correct order:

How electricity flows from power stations to our homes, schools, and offices.	Order	Order
1. Substations reduce high-voltage electricity to a lower voltage, which can be used in homes and offices.	6	3
2. An electrical appliance draws the energy needed to work from the mains supply.	3	7
3. Powerlines carry electricity from the power stations to substations in areas where people live and work.	1	2
4. Wall sockets built into the walls of a building provide places for people to safely plug into the mains supply.	7	5
5. When plugged into the wall socket, an appliance is connected to the mains electrical supply.	4	6
6. Power stations burn fuels such as coal to generate electricity.	5	1
7. Electricity boxes at home control the supply of electricity to the building.	2	4

Question 7: Systems for moving things (Distinguish)

Examine the following images and answer the questions that follow.

7.1 How would the following system make a person's life easier. **/3**



a) When gardening no need to carry everything. Wheelbarrow can be pushed around effortlessly. Making the movement of objects easier.

b) Heavy duty trolleys can be useful for all sorts of objects - and for items of all shapes and sizes. They can help load and unload items safely, as well as store and transport goods.

c) These not only offer a pleasing design, but also excellent suspension and stability. The large wheels allow the pram to glide effortlessly over bumps and obstacles and provide your child with a comfortable riding experience.

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SECTION B – Planet Earth **/ 19**

Question 1:

State whether the following sentences are true or false. **/11**
If false, rewrite the statement to make it true.

a) The Earth's axis is vertical. False

The Earth's axis is slightly tilted.

b) The whole Earth can receive light at the same time. False

Only the part of the Earth facing the sun can receive light at a given time.

c) The side of the Earth that receives light is experiencing night. False

The side of the Earth that receives light is experiencing daylight.

d) The Earth takes 24 hours to complete one rotation on its axis. True

e) The Sun takes 365 ¼ days to revolve around the Earth. False

The Earth takes 365 ¼ days to revolve around the Sun.

f) The shape of the Earth's orbit is square. False

The shape of the Earth's orbit is oval or sphere.

Question 2: **/8**

2.1 Write a full sentence explaining **where soil comes from.** **/1**

Soil comes from rocks. Soil is rock that has broken down into small particles. Different types of rock form different types of soil.

2.2 Write a full sentence explaining **what soil is made up of.** **/4**

Weathered or broken-down rock.
Remains of dead organisms. - Humus (decaying plant and animal matter)
Air
Water

2.3 Complete the following table by filling in the missing information. **/3**

	Clay	**Loam**	**Sand**
Size of particles	Most of the particles in clayey soil are small	Loam contains sand and clay particles; mixed sizes.	Most of the particles in sandy soil are quite large
Drainage	Clayey soil holds a lot of water	Loam soil lets some water pass through	Sandy soil does not hold water well.

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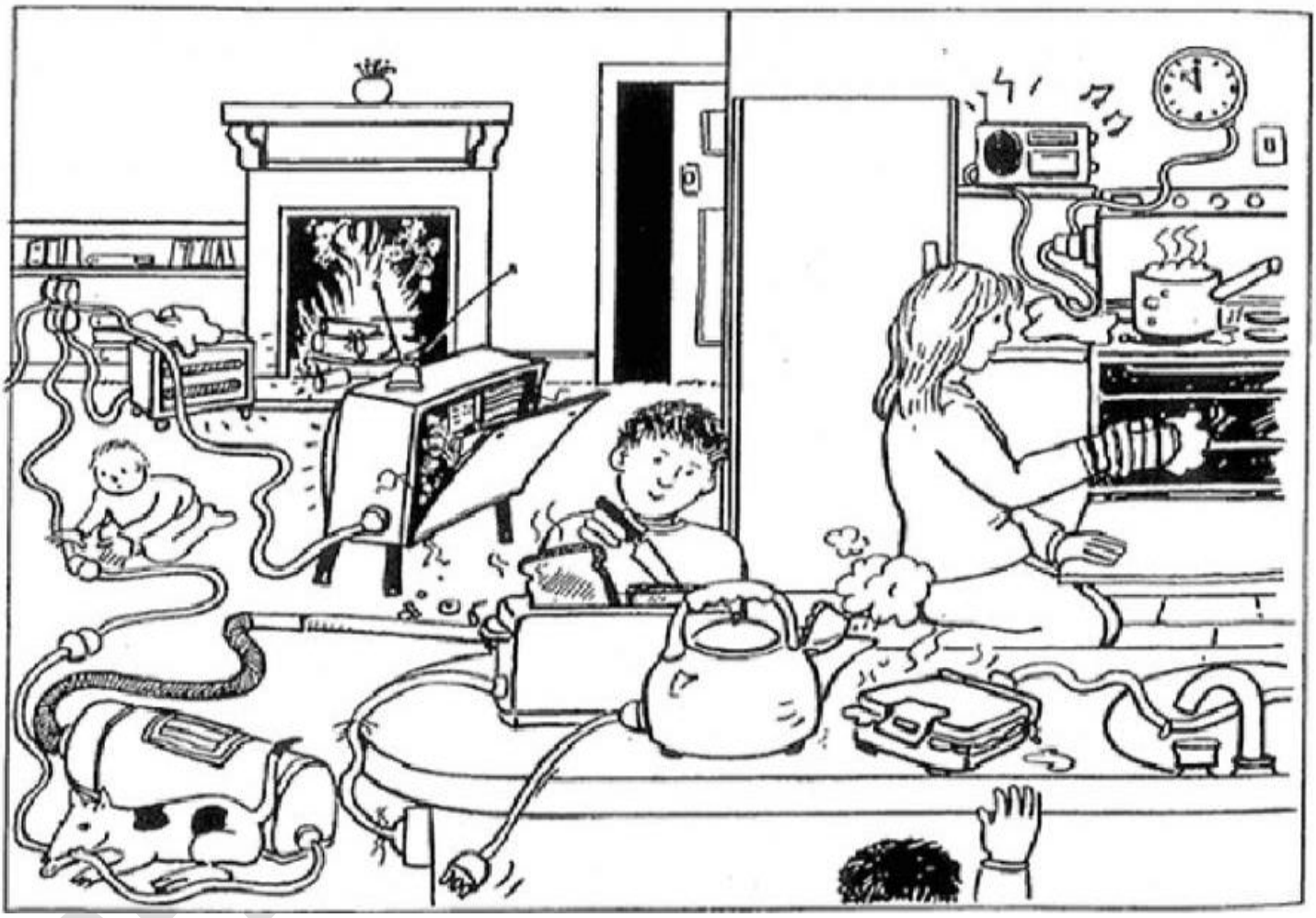
Nutrients	Clayey soil has some nutrients	Loam contains the right amount of nutrients	Sandy soil contains very few nutrients

Extension Activity

To be completed after you have checked **all your answers** and only if you have finished.

Spot the Electrical Hazards